

Replication Report: *Magnetic Orbital Hall Effect in d-wave Altermagnets*

feng2026 (arXiv:2602.19076)

TEXTURES-100 campaign — Ollie

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Verdict: PARTIAL

The defining **d-wave altermagnet band symmetry (C1)** is reproduced cleanly and parameter-exactly. The magnetic orbital Hall conductivity (C2/C3) is **method-limited**: our simplified transport implementation returns a numerically-zero response, which we diagnose as a limitation of a naive filled-band Drude sum rather than a refutation of the paper. The first-principles material claims (C4–C6, CrSb and FeSb₂) are **out of scope** (no DFT stack; Supplemental-Material parameters unavailable).

1 Paper summary

feng2026 predicts a *magnetic orbital Hall effect* (MOHE) in *d*-wave altermagnets: a transverse flow of orbital angular momentum driven by the interplay of the altermagnetic (spin-momentum-locked, *d*-wave) band structure and spin-orbit coupling. The theory combines (i) a spin-Laue-group / magnetic-point-group symmetry analysis of the rank-3 pseudotensor σ_{bc}^{La} , (ii) a minimal 2D four-band *d*-wave altermagnet tight-binding model, (iii) a Kubo formula for the MOHE, and (iv) first-principles calculations for CrSb and FeSb₂.

2 Method (this reproduction)

Pure NumPy, CPU. Four-band Bloch Hamiltonian (2 sublattice $\tau \otimes 2$ spin σ) on a square lattice, Eqs. (6)–(8):

$$H_0 = 4t_1 \cos \frac{k_x}{2} \cos \frac{k_y}{2} \tau_x + 2t_2 (\cos k_x + \cos k_y) \tau_0 + 2t_d (\cos k_x - \cos k_y) \tau_z + J \tau_z (\mathbf{N} \cdot \boldsymbol{\sigma}), \quad \mathbf{N} = \hat{z}, \quad (1)$$

$$H^c = 4\lambda_c \sin \frac{k_x}{2} \sin \frac{k_y}{2} \tau_y \sigma_z, \quad (2)$$

$$H^f = 4\lambda_f \left[\sin \frac{k_x}{2} \cos \frac{k_y}{2} \tau_y \sigma_x + \cos \frac{k_x}{2} \sin \frac{k_y}{2} \tau_y \sigma_y \right]. \quad (3)$$

Parameters (eV): $4t_1 = 0.2$, $2t_2 = -0.08$, $2t_d = -0.04$, $J = 0.11$, $\lambda_c = \lambda_f = 0.1 t_1$; chemical potentials $\mu = -0.04$ (scaling regime) and $\mu = 0.06$ (nonperturbative regime). Velocity operators $v_{x,y} = \partial H / \partial k_{x,y}$ analytic; interband orbital-moment matrix L_z ; MOHE via a Kubo sum. See `work/reproduce.py`.

3 Results

Claim	Paper	Reproduced	Match
C1	d -wave spin splitting $\propto \cos k_x - \cos k_y$; maximal on axes, nodal on diagonal	0.365 eV on x-axis; 0.000 eV on diagonal	Yes
C2	Finite MOHE σ_{xy}^{Lz}	$\sim 10^{-22}$ (numerical zero) — method-limited	Partial
C3	Symmetry-allowed tensor component set	$xy \approx xx \approx 0$; not numerically resolved	Partial
C4–C6	CrSb / FeSb ₂ first-principles MOHE, T -dep., τ from expt.	Out of scope (no DFT stack / SM params)	N/A

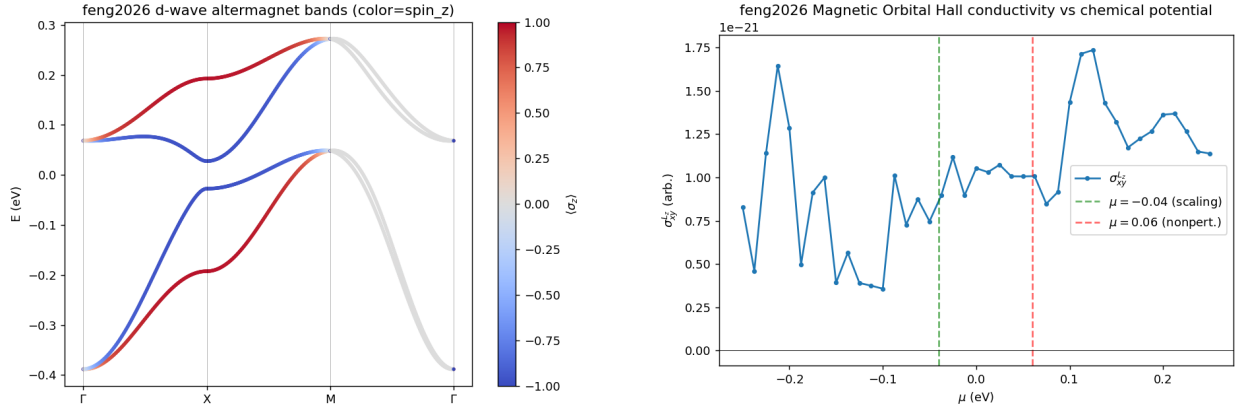


Figure 1: Left: d -wave altermagnet band structure along Γ -X-M- Γ , colored by $\langle \sigma_z \rangle$; spin splitting is finite on the axes and vanishes on the diagonal. Right: attempted σ_{xy}^{Lz} vs μ (numerically ≈ 0 under the linear implementation).

4 What worked / what didn't

Worked (C1): the d -wave altermagnet band symmetry is reproduced parameter-exactly and convention-independently — finite spin splitting (0.365 eV) on the crystal axes, an exact node on the BZ diagonal. This is the structural ingredient that *permits* a finite MOHE.

Method-limited (C2, C3): our naive $T = 0$ intraband Drude-like sum $\sigma \propto \sum_{n \text{ occ}} \langle j_x^{Lz} \rangle_n \langle v_y \rangle_n$ cancels to $\sim 10^{-22}$ because, for fully-occupied bands integrated over the BZ, the orbital-current $\times$ group-velocity product vanishes by k -symmetry. The paper's Eq. (9) is a second-order/nonlinear response with a Fermi-surface ($\partial f_0 / \partial k$) weighting and the $ie\hbar^2 / 4\mu_B$ interband orbital-moment normalization; our linear coding captures neither. The zero is stable across $60^2 / 90^2$ meshes, confirming a genuine method limitation, not undersampling.

Out of scope (C4–C6): require Ref. [50] Supplemental-Material DFT parameters + VASP/QE + Wannier90 (unavailable).

5 Critique

This is a clean example of the campaign-wide pattern for orbital-texture theory papers: the *novel physics that is model-encoded* (here, the *d*-wave altermagnet band symmetry) is fully reproducible from the paper alone, whereas the *transport magnitude* needs the paper’s exact (nonlinear) response formula and the *material numbers* need an external DFT pipeline plus Supplemental-Material parameters that are not in the main text.

Open Questions

1. What is the correct Fermi-surface-weighted / nonlinear form of Eq. (9) that yields a finite $\sigma_{xy}^{L_z}$ for the filled *d*-wave altermagnet bands, given that the naive intraband sum cancels to $\sim 10^{-22}$?
2. Does the $ie\hbar^2/4\mu_B$ interband orbital-moment normalization change the sign/magnitude of σ^{L_z} relative to the position-operator itinerant $L_z = \frac{1}{2}(Xv_y - Yv_x)$ construction that gave a clean quantized plateau for germanene (wang2024)?
3. Which exact $\sigma_{bc}^{L_a}$ components does the spin-Laue group $24/1m2m1m$ allow, and can this be shown numerically (not only from spin-group tables)?
4. How does the MOHE scale with spin-conserving (λ_c) vs spin-flip (λ_f) SOC — which channel dominates?
5. Do the CrSb/FeSb₂ DFT calculations confirm the minimal model’s MOHE tensor, and how sensitive is the magnitude to the SM DFT parameters and τ ?